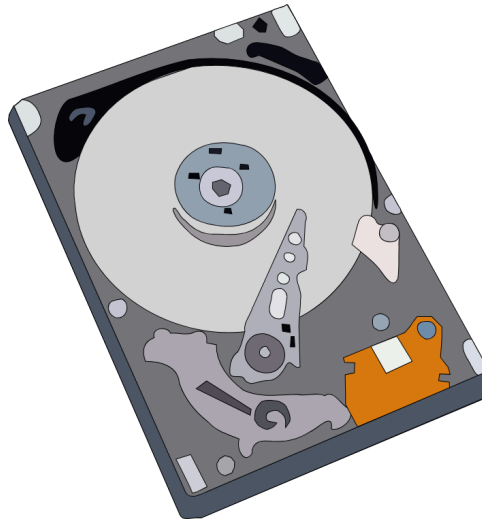


Network-Attached_Storage_Notes



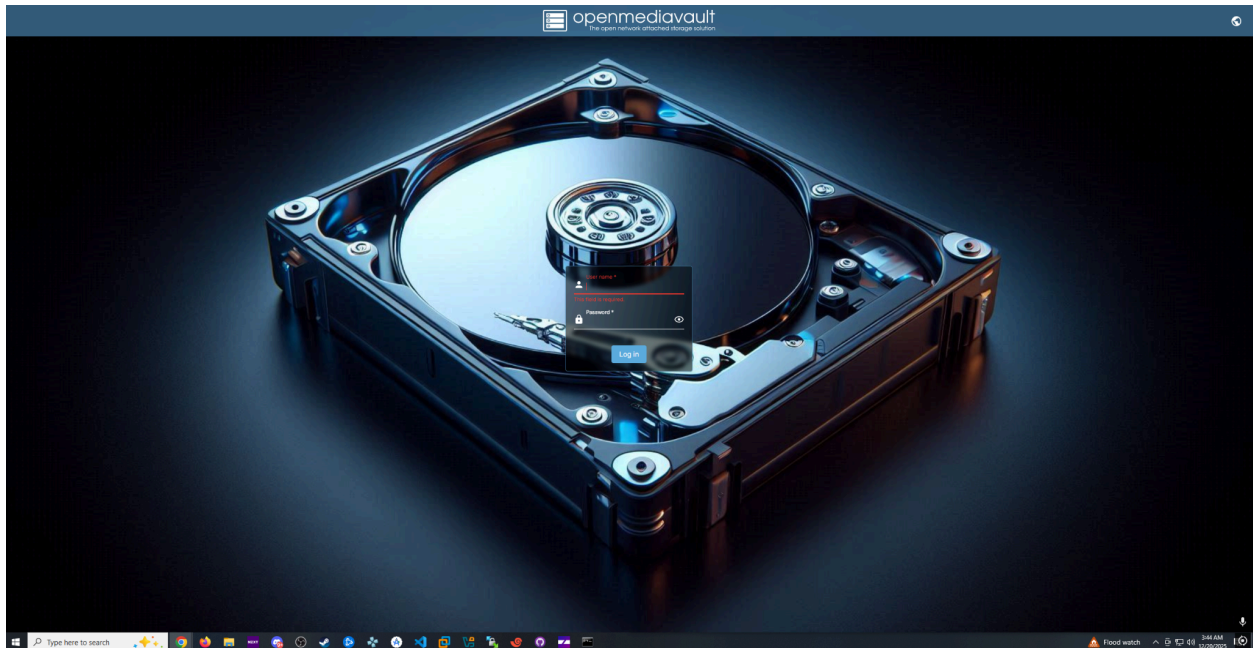
NAS will be used as a shared home network drive for convenience use. Pictures, saved short video clips moments, videos, documents! This will be done with OpenMediaVault (OMV) - a web-based OS I install on my Pi to turn it into a NAS. A home file server. It does control permissions via users. OMV configures **Samba**, a Linux File-sharing service that listens on the network and accepts requests from other computers on the HDD and enforces the permissions.

Hardware Required

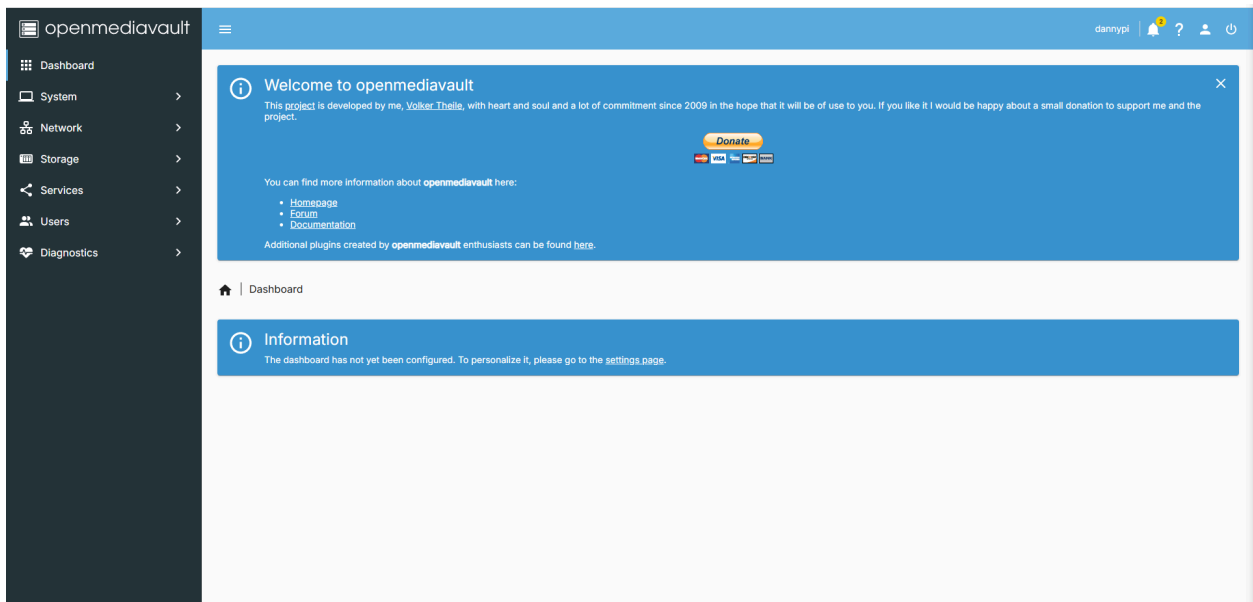
- Sabrent USB 3.0 3.5 HDD enclosure
 - <https://a.co/d/4StL0Kf>
- Seagate BarraCuda 2TB 3.5 HDD
 - <https://a.co/d/gE122cM>

To open OpenMediaVault (OMV), goto the Pi's IP in any device browser
OR

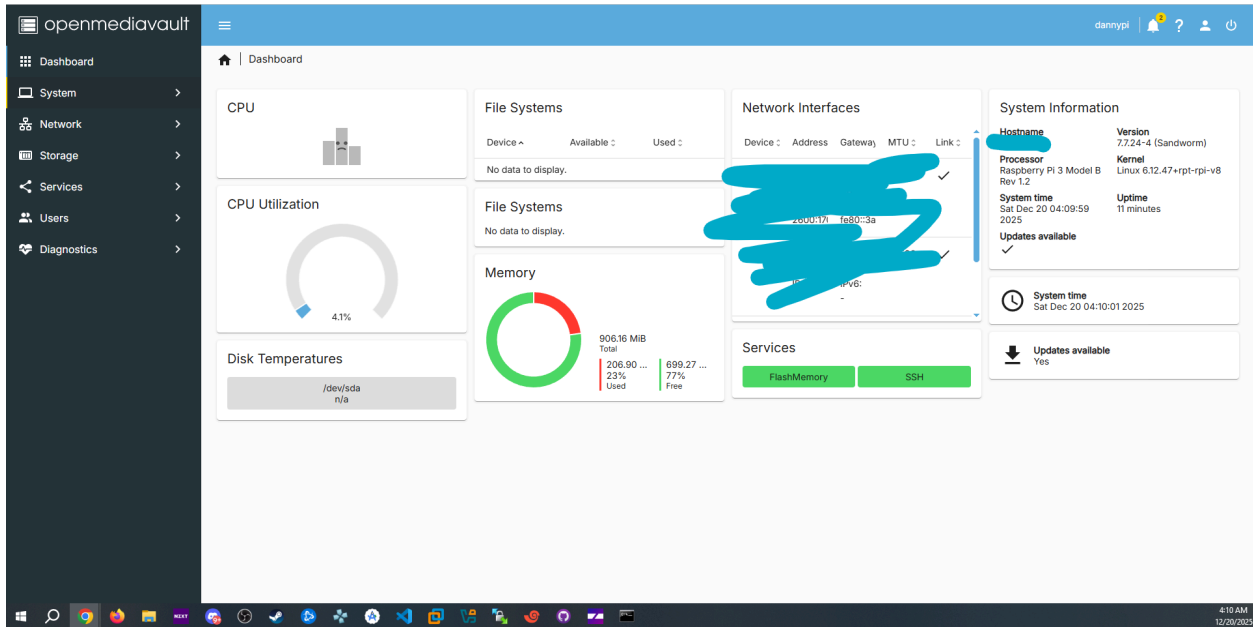
The same thing as ^, do <hostname.local> so you don't have to get the Pi IP



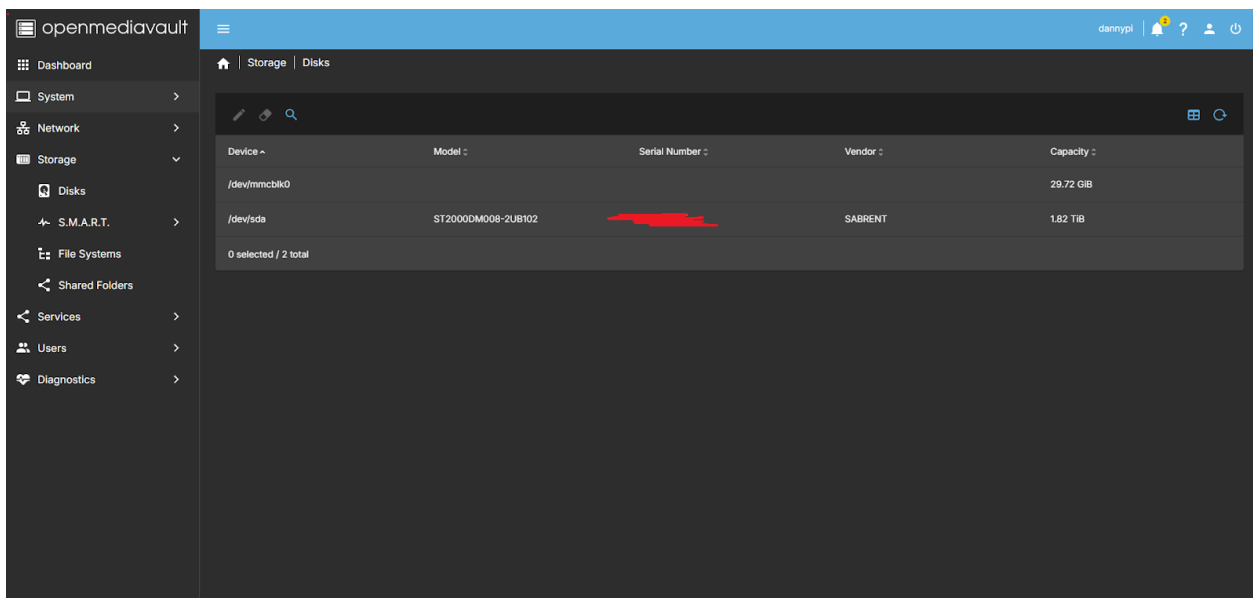
OMV after using <hostname.local>



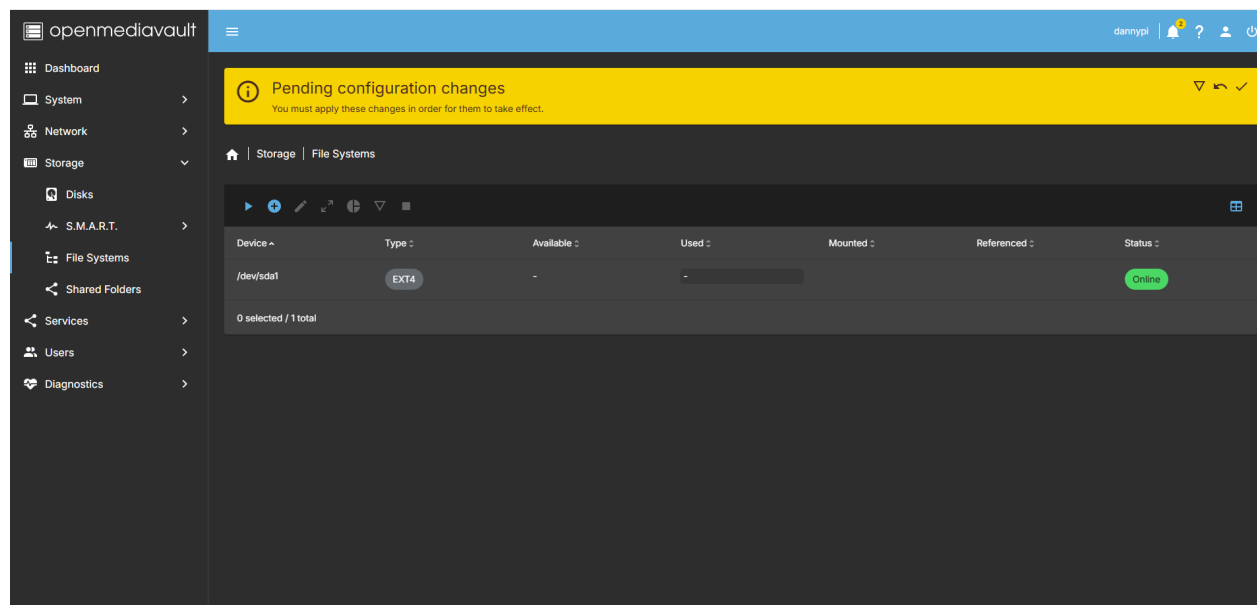
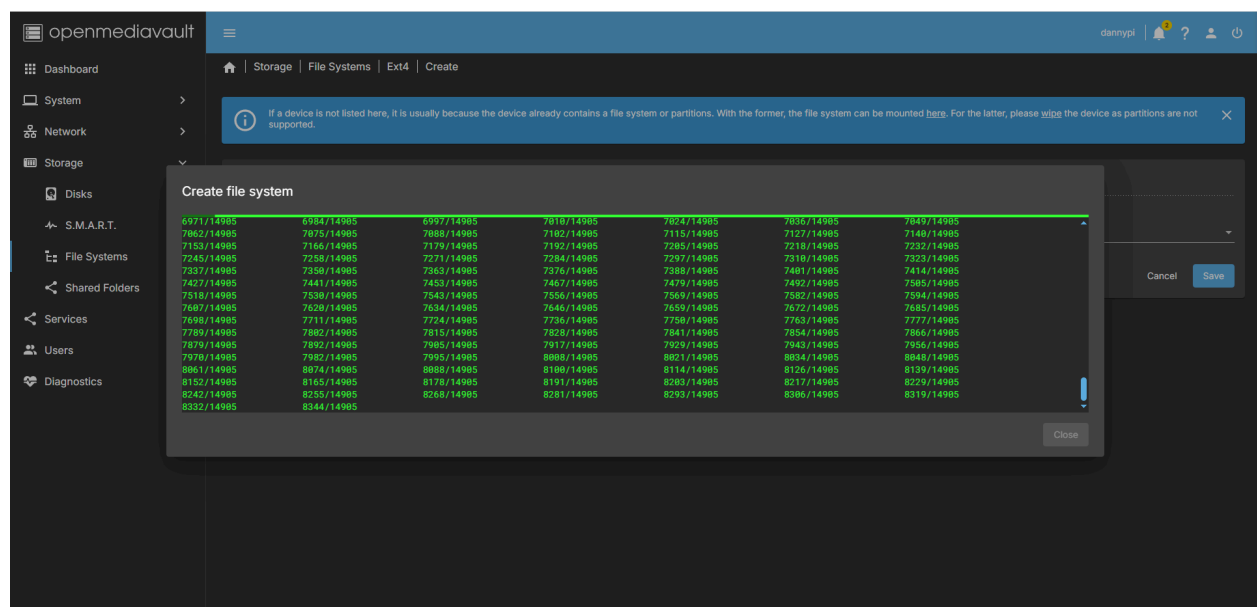
Post admin login (details in Secret Notes)D



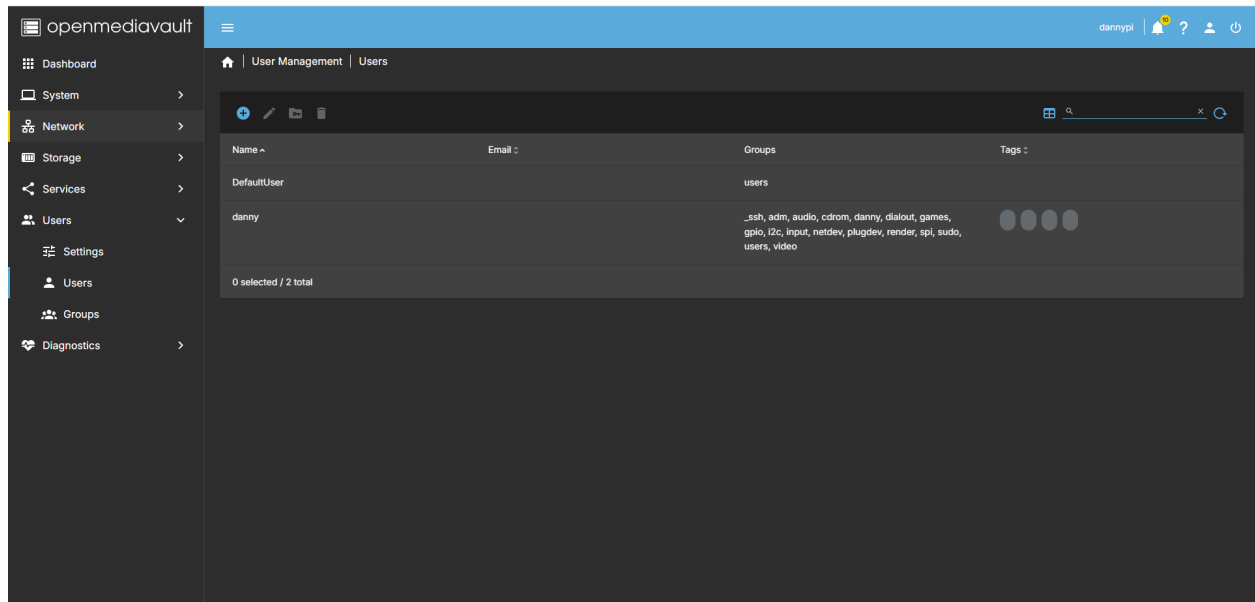
Populating OMV dashboard with settings in the UI



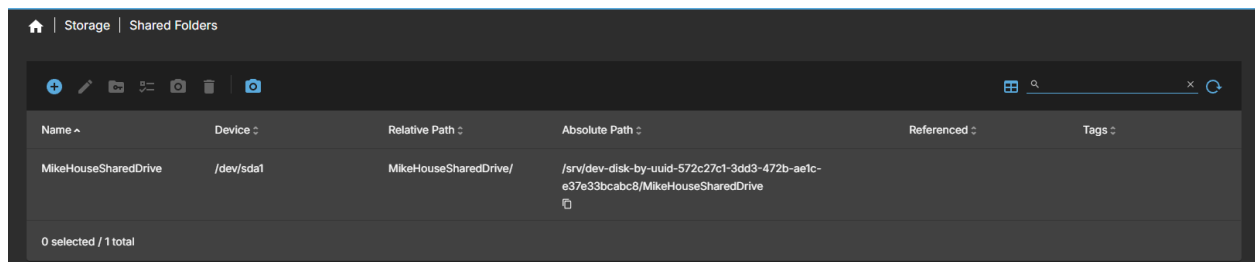
Going to Storage > Disk and I see my 2TB HDD



Adding file system to drive

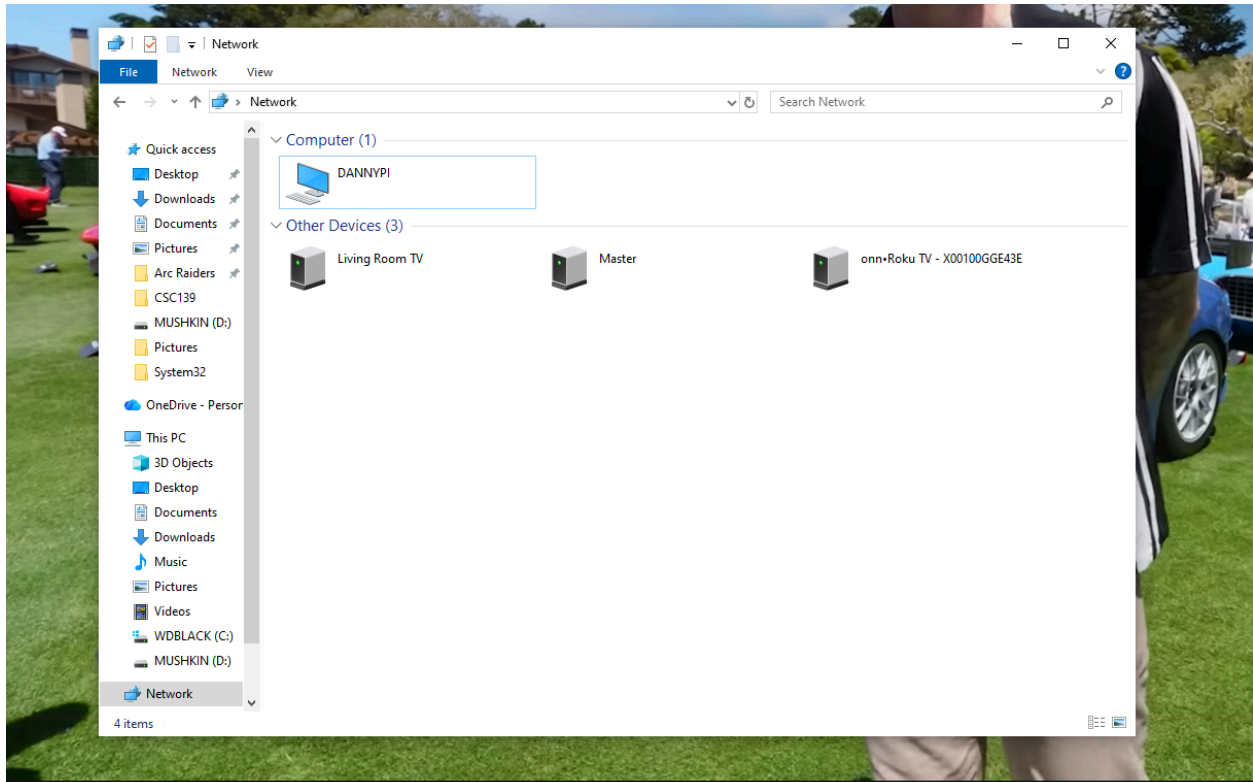


Adding a User **(This is where you make logins for connecting to the drive)**



Adding Shared Folder

Added Recycling bin >:)



Monterey Car Week BROKE My Brain



AlexMartini
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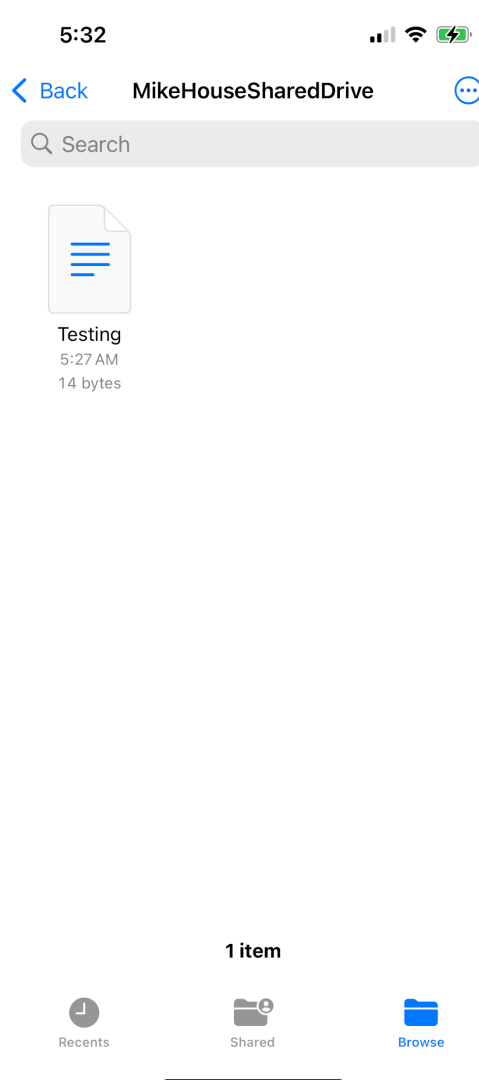
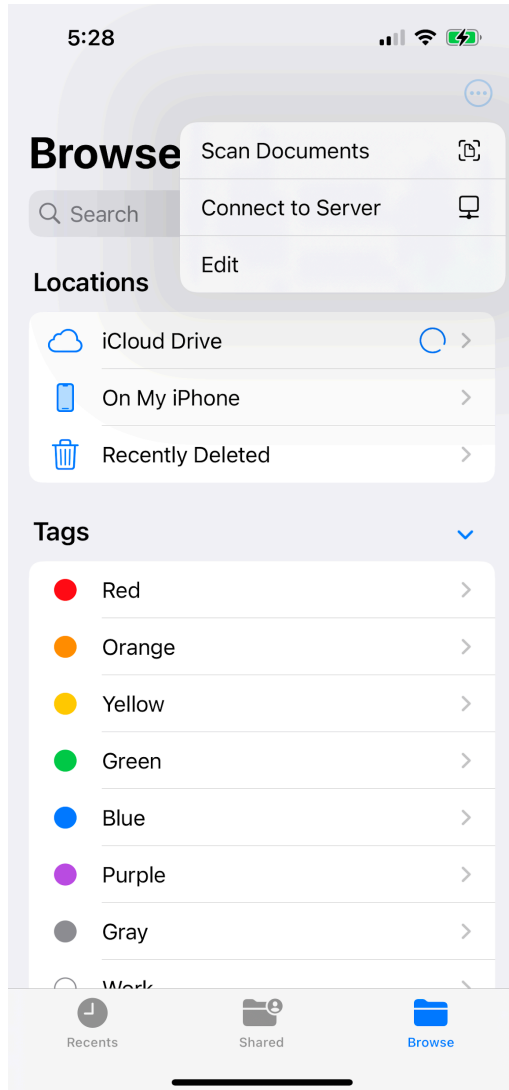
Share

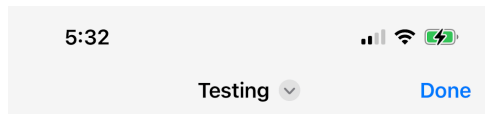
Ask

Save



NAS POPS UP ON MY NETWORK AFTER ENABLING SMB!





Testing file



To connect to NAS via iPhone

File > 3 dots top right > Connect to Server smb://<pi_ip>

IT WORKED and the file I created, I was able to access!

To make a folder that only 1 or more people can see

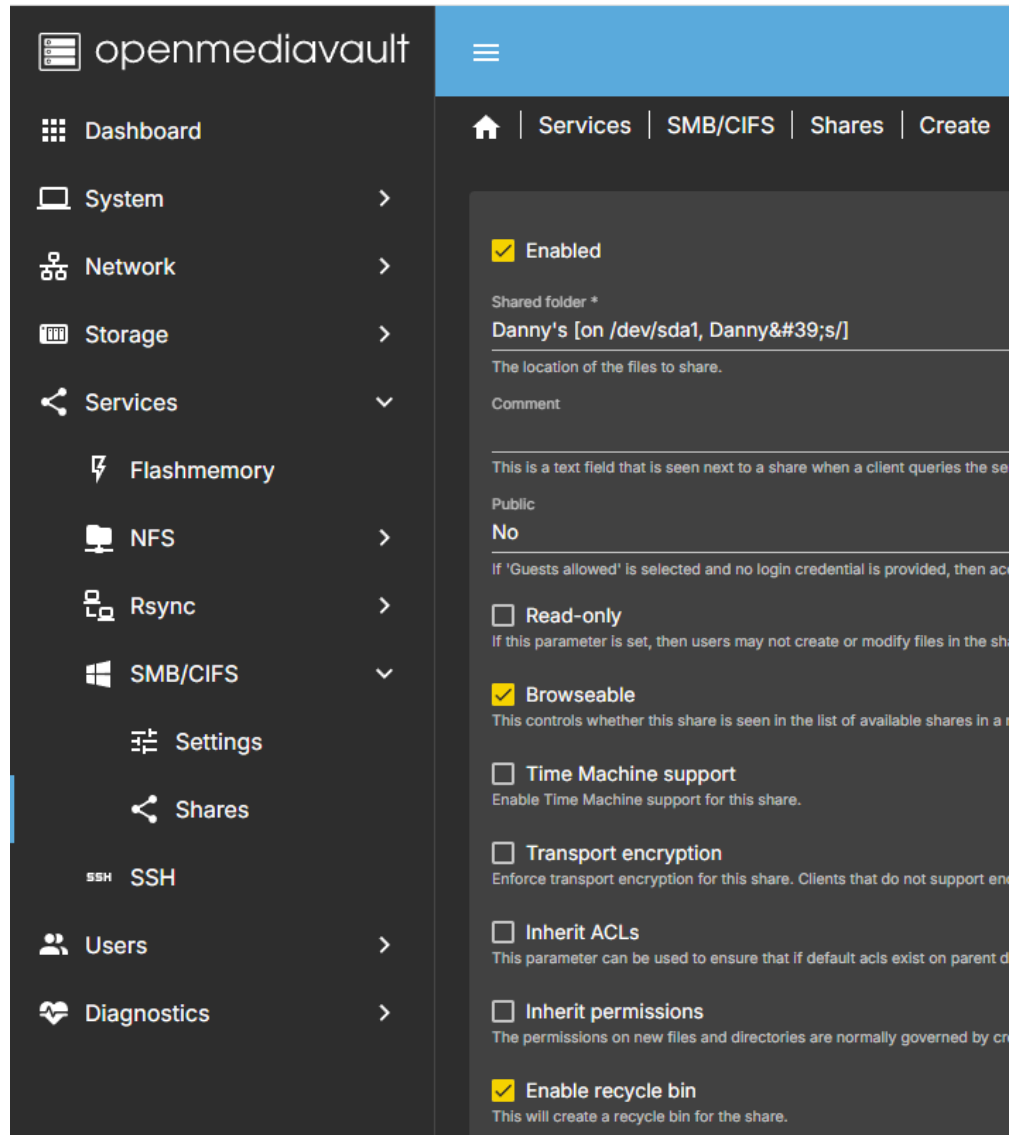
1. Goto Storage > Shared Folders
2. + Create

The screenshot displays the OpenStack dashboard interface. On the left, a sidebar menu contains the following items: Dashboard, System, Network, Storage (expanded), Disks, S.M.A.R.T., File Systems, Shared Folders (highlighted), Services, Users, and Diagnostics. The main panel on the right is titled 'Create' and contains a form for creating a new shared folder. The form fields are as follows:

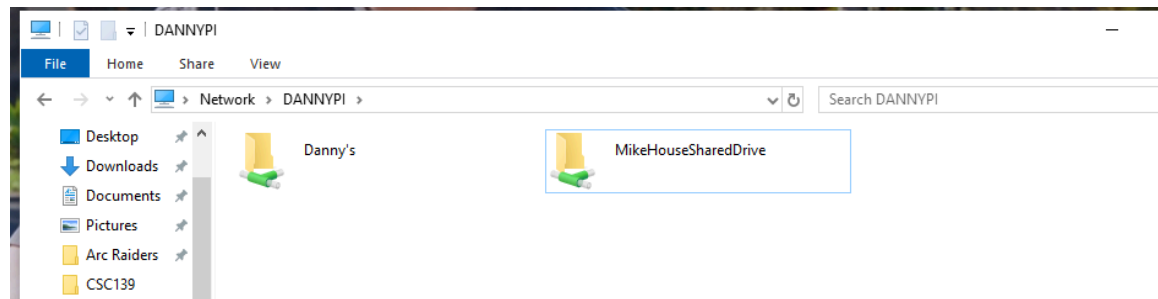
- Name ***: example
- File system ***: /dev/sda1 [EXT4, 60.00 KiB (1%) used, 1.78 TiB available]
The file system on which the shared folder is to be created.
- Relative path ***: example/
The relative path of the folder to share. The specified folder will be created if it does not exist.
- Permissions ***: Administrator: read/write, Users: read/write, Others: read-only
The file mode of the shared folder path.
- Tags**: (empty field)

a.

3. Services > SMB/CIFS
4. + Create and fill out using new folder created
 - a. Recommend Enabling Recycling bin



b.



c.

It works!

Storage | Shared Folders | ACL

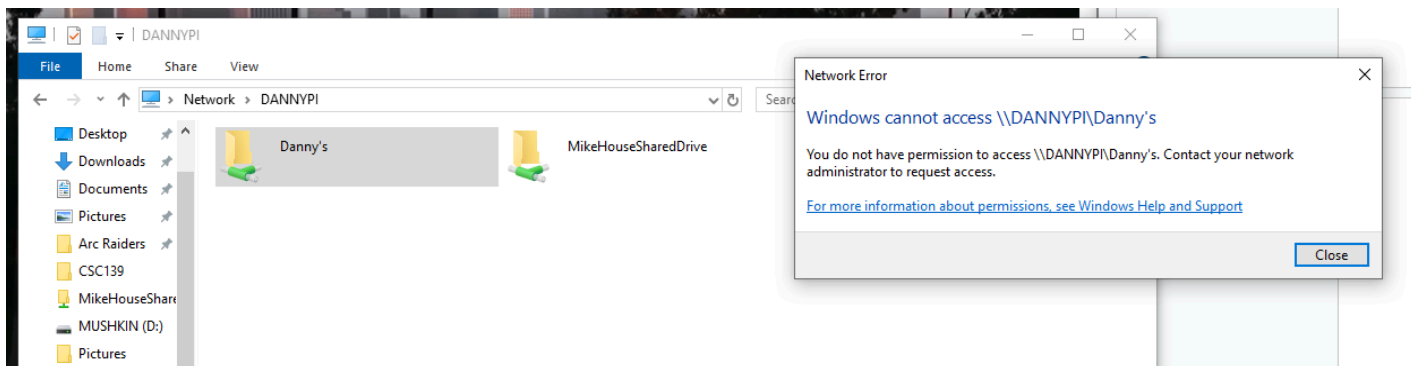
Permissions of others (e.g. anonymous FTP users).

File access control lists

User/Group permissions

Name ^	ID ^	Type ^	System account ^	Permissions ^
danny	1000	User		Read/Write Read-only No access
danny	1000	Group		Read/Write Read-only No access
DannyActual	1002	User		Read/Write Read-only No access
DefaultUser	1001	User		Read/Write Read-only No access
..apt	42	User	✓	Read/Write Read-only No access

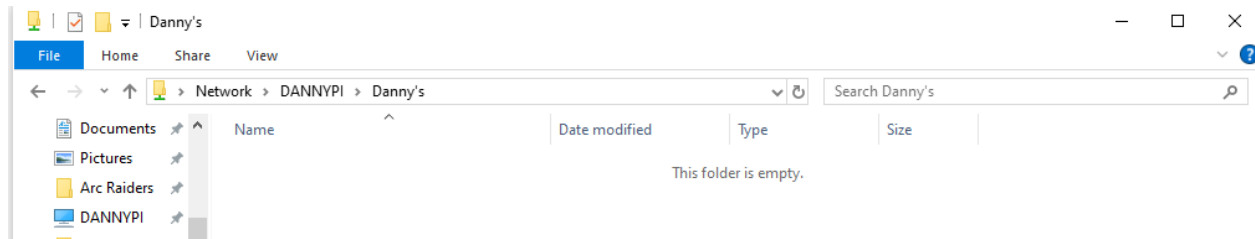
To decide what users have access to a Shared Folder, goto Storage > Shared Folders



Can't open since the only user with access is DannyActual and I'm logged in with DefaultUser here.

To switch users

1. Close File Explorer
2. Command Prompt `net use * /delete` and next time it will ask you for credentials. **Sometimes it takes retries or time**



I swapped to DannyActual user here which has permissions to read/write into that Shared folder

In case you forgot admin login (Yeah this happened to me)

`sudo omv-firstaid`

Please note that I did consider having another 2TB drive as a 1-to-1 mirror backup of the first 2TB drive. No raid setup as this will work better for my setup. However I do not have the funds to do so and don't personally think I need it. I can attach a small 128gb external SSD that connects via USB down the line for only about \$25 for those most important and sensitive documents.

01/06/2025

Brother said I was crazy thinking we'd only use 2GT so he provided me with a 6TB HDD to replace the 2TB. Setup was simple, I just referred back to these notes, starting at where we used OpenMediaVault (OMV) to set up the drive's file system. Personally I will be using about the same amount of storage once I have my own home under my name, but with 1-to-1 mirror for backup. Raid does not seem to give me any advantages over this since I will simply only have two physical drives